Ojas Mediratta

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Education

Georgia Institute of Technology — Atlanta, GA
M.S. Robotics | *Specialization in Artificial Intelligence, Perception, and Mechanics* — *Expected May 2027*

Georgia Institute of Technology — Atlanta, GA
B.S. Computer Engineering | *Graduated with High Honors* — *May 2025*

Experience

Robotics Hardware Intern — Aug 2026 – Dec 2026 (Incoming)
Physical Intelligence (π) — *San Francisco, CA*

- Incoming member of the hardware engineering team developing general-purpose robotic systems

Robotics Research Intern — May 2026 – Aug 2026
GE Vernova Advanced Research — *Niskayuna, NY*

- Led end-to-end integration and validation of a marine robot for confined-space inspection, spanning hardware, electronics, embedded software, and field-ready operation
- Developed a Linux-based control and telemetry stack integrating motors, onboard sensors, and tethered data streaming for remote inspection and maintenance data collection with BlueOS
- Operationalized newly-acquired Voltera PCB fabrication systems and authored standard operating procedures for safe, repeatable in-house PCB prototyping

Graduate Research Assistant — Jan 2026 – Present
Georgia Institute of Technology - Laboratory for Intelligent Decision and Autonomous Robots — *Atlanta, GA*

- Designed, fabricated, and characterized flexible tactile sensor arrays for humanoid end effectors, forearms, and torso, increasing sensing coverage and spatial resolution across curved body surfaces
- Designed and fabricated wearable harnesses and sensor mounts that enabled repeatable sensor placement during human data collection and real-time tactile sensing

Projects

Cetacean Research ROV | *C++, ESP32, Raspberry Pi, Python, Fusion, KiCAD* — Aug 2024 – May 2026

- Built a remotely operated vehicle for dolphin research and enrichment, contributing across firmware, electronics, controls, and mechanical design; validated the system through 15+ pool trials and four open-ocean deployments
- Developed ESP32 firmware integrating cascaded depth and pitch PID controllers, ESC-driven thrusters, onboard sensors, over-the-air telemetry, and status signaling, maintaining less than 5° steady-state attitude error in water
- Built a real-time DSP pipeline on Raspberry Pi using filtering, autocorrelation, and Goertzel detection on hydrophone audio, converting dolphin vocalizations into robot commands with 300-500 ms actuation latency
- Designed PCBs using KiCAD, integrating microcontroller, regulated power, and sensor interfaces, reducing point-to-point wiring, assembly complexity, and potential field failure points

MineTrack Visual Odometry | *Python, PyTorch, Machine Learning, Computer Vision* — Jan 2026 – May 2026

- Developed a machine learning-based visual odometry pipeline that estimated 6-DoF camera motion from sequential Minecraft game play frames and reconstructed full 3D trajectories
- Trained CNN-MLP, LSTM, and transformer architectures for relative translation and rotation prediction
- Implemented a dual-autoencoder that learned compact representations of scene content and inter-frame motion
- Built evaluation tools for trajectory error, endpoint drift, yaw tracking, and path-length accuracy

TurtleBot3 Autonomy | *ROS2, Python, OpenCV, Gazebo, Control, Motion Planning* — Aug 2025 – Dec 2025

- Developed a ROS2 vision pipeline with OpenCV for real-time object tracking and following with >95% success
- Designed and tuned PID controllers for differential-drive motion, reducing steady-state error by 35%
- Programmed grid and probabilistic path planners with python and ROS2, in a multi-node architecture, raising navigation success from 60% to 95% and eliminating collisions across multi-waypoint maze runs
- Fused odometry and sensor data, maintaining <10 cm localization error over runs with moving obstacles

Skills

Software: C, C++, Java, MATLAB, Python, Pandas, Pytorch, ROS2, Android, Kotlin
Hardware: Arduino, Raspberry Pi, ESP32, ARM, RISC-V
Protocols: TCP/IP, I2C, CAN, UART, SPI, Serial, USB, TTL, SBUS
Developer Tools: VSCode, Arduino IDE, Android Studio, Fusion, Gazebo, KiCAD, Git, Docker
Lab Tools: Oscilloscope, Multimeter, Soldering, 3D Printing, CNC Mill, Laser Cutter, Logic Analyzer
Advanced Topics: Firmware Programming, Electronics, Computer Networking, PCB Design, Mechanical Design